

Register No:

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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110.

(An Autonomous Institution, Affiliated to Anna University, Chennai)

M.Tech. CSE (5YI) End Semester Theory Examinations, November / December 2025.

Fifth Semester

Computer Science and Engineering

ICS1503 WEB APPLICATION DEVELOPMENT

(Regulations 2023)

Time: Three Hours

Answer ALL Questions

Maximum:100 Marks

K1: Remembering

K2: Understanding

K3: Applying

K4: Analyzing

K5: Evaluating

K6: Creating

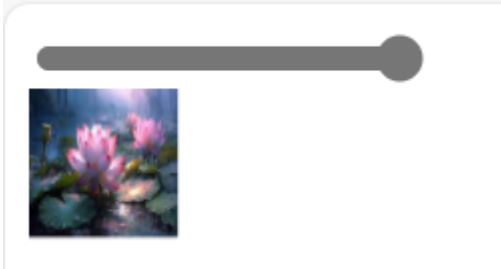
CO1:	Make use of HTML5 and CSS3 to design modern web site (K3)
CO2:	Utilize Javascript and DOM to implement dynamic web page (K3)
CO3:	Develop web applications using Servlets (K3)
CO4:	Develop micro services using Spring Boot (K3)
CO5:	Develop full stack web applications using Vue JS and Spring Boot (K6)

Part – A (5 × 2 = 10 Marks)

		KL	CO	PI
1.	Explain difference between cell padding and cell spacing.	K2	CO1	1.4.1
2.	Define an anonymous function in JavaScript with an example.	K1	CO2	1.4.1
3.	What is the use of the sendRedirect() method in Servlets?	K1	CO3	1.4.1
4.	Explain @RestController with an example.	K2	CO4	1.4.1
5.	List out the features of Vue.js.	K1	CO5	1.4.1

Part – B (5 × 6 = 30 Marks)

		KL	CO	PI
6.	Develop an HTML & CSS program to create three boxes with different border colors: Box 1 (red border), Box 2 (green border), and Box 3 (blue border). Each box should use box-sizing: content-box and have the following properties: border of 5px, padding of 6px and 7px, margin of 16px,15px, and 17px, with a content width of 50px and	K3	CO1	1.4.1 2.1.2

	height of 50px. Arrange the boxes in a row on large screens and stack them vertically on small screens (responsive design). calculate the overall size of a single box.			
7.	Construct a JavaScript program using DOM manipulation to change the displayed image dynamically based on the value selected from a slider. The slider should have multiple ranges, and moving the slider should update the image accordingly. 	K3	CO2	1.4.1 2.1.2
8.	Explain in detail how servlet generates dynamic response based on the client request with suitable diagrams.	K2	CO3	1.4.1
9.	Explain the architecture of Model–View–Controller (MVC), with its end-to-end flow of request and response by considering a simple example in Spring Boot.	K2	CO4	1.4.1 2.1.2 3.4.2
10.	Identify and briefly explain the output of the following Vue.js functional component: <pre> <template> <div> <button @click="toggleShow"> {{ show ? "Hide Element Below" : "Show Element Below" }} </button> <div v-if="show">Toggle Challenge</div> </div> </template> <script> export default { name: "App", data() { return { show: true </pre>	K3	CO5	1.4.1 2.1.2

	<pre> }; }, methods: { toggleShow() { this.show = !this.show; } } }; </script> </pre>			
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Part – C (5 × 12 = 60 Marks)

		KL	CO	PI
11.	Explain the commonly used HTML form elements, their attributes and functions, with an example code for a Visa Application Registration Form. The form should include the essential fields required for visa registration such as Full Name, Passport Number, Email, Date of Birth, Gender, Nationality, Visa Type, Address, Travel Dates, Upload Passport Copy, Declaration Checkbox, and Submit Button.	K2	CO1	2.1.2 3.4.2 13.3.1
(Or)				
12.	Explain in detail the process of CSS Selectors with suitable examples.	K2	CO1	2.1.2 3.4.2 13.3.1
(Or)				
13.	(a) Develop a program in JavaScript to display the age of the person, given the date of birth. [7 Marks] (b) Develop a program that dynamically creates a paragraph element and changes the background color of the third and fourth list items when the mouse hovers over the newly created paragraph element. [Assume total 5 paragraph elements are available]. [5 Marks]	K3	CO2	2.1.2 3.4.2 13.3.1
(Or)				
14.	(a) Develop JavaScript code to do the following: i. Insert a paragraph element containing the text “WAD ”	K3	CO2	2.1.2 3.4.2 13.3.1

	ii. Change content of the paragraph element by “ICS3102-WAD” Iii. Change the background color of the text in paragraph element iv. Change the color of the text in paragraph element [7 Marks] (b) Add a new element containing the faculty handling the course when the web page is loaded itself and inspect the names of the node objects present in the DOM tree. [5 Marks]																			
15.	Develop a Currency Converter Web Application with the following interface. The currency values to be used are the Indian rupee, US dollar, euro, and pound. Given a value, the source currency and the currency to be converted, an equivalent currency value should be displayed in the text box. - Create a simple interface to get the input and display the output. - Create a table called CurrencyValues in MySQL to have the following: <table><tr><td>Rupee</td><td>Euro</td><td>Dollar</td><td>Pound</td></tr><tr><td>1</td><td>0.13</td><td>0.014</td><td>0.011</td></tr><tr><td>8</td><td>1.20</td><td>1.09</td><td>0.8</td></tr><tr><td>7</td><td>1.25</td><td>0.92</td><td>0.77</td></tr></table> - Write a server that queries the database to display the destination currency value for the given source currency value when the “GO” button is clicked.	Rupee	Euro	Dollar	Pound	1	0.13	0.014	0.011	8	1.20	1.09	0.8	7	1.25	0.92	0.77	K3	CO3	2.1.2 3.4.2 13.3.1
Rupee	Euro	Dollar	Pound																	
1	0.13	0.014	0.011																	
8	1.20	1.09	0.8																	
7	1.25	0.92	0.77																	
16.	Develop a web application that collects personal information across multiple pages with session tracking using cookies and displays it on the final webpage. The scenario is as follows: On the first webpage, get the Name and ID number as input from the user.	K3	CO3	2.1.2 3.4.2 13.3.1																

	- On the second webpage, get the Phone Number as input from the user. - On the third webpage, get the Address as input from the user. - On the fourth webpage, display all the information collected from the previous pages (Name, ID number, Phone Number, and Address) using cookies.			
17.	Develop a Travel Guide backend application using Spring Boot with MongoDB. The application should include a City entity, a repository interface to interact with the database, and controllers with endpoints to execute the following: retrieving all cities, retrieving hot cities, and retrieving cities with more than five historical places. The endpoints should accept and return data in JSON format and can be tested using Postman. Ensure that the application is structured with proper layers (controller, service, repository) to manage business logic and database interactions efficiently.	K3	CO4	2.1.2 3.4.2 13.3.1
(Or)				
18.	Develop a backend Grocery application using Spring Boot and MongoDB. The application should include: - Entity representing the Grocery collection. - Repository Layer to interact with the database. - Service Layer to interact with the Repository layer. - Controllers with the following REST endpoints: - POST/grocery-Store all grocery items - GET /grocery/all – Retrieve all items - GET /grocery/same-price – Retrieve grocery items with same prices - DELETE /grocery/{itemId}	K4	CO4	2.1.2 3.4.2 13.3.1
19.	Develop a Front-end application using Vue.js to create a box container containing n number of boxes. Each box should have a random background color when rendered.	K3	CO5	2.1.2 3.4.2 13.3.1

	After rendering, when a box is clicked, its background color should change to a different color. If the newly generated random color is the same as the current color, the box's background color should be set to white.			
(Or)				
20.	Develop a Vue.js functional component called ColorPalette to change the background color of a text box based on the selected color as shown in below figure. Use a reactive state variable to keep track of the chosen color. If there are '<i>n</i>' colors in the palette, there should be '<i>n</i>' actions to update the color. Refactor the logic using Vue's reactive state and a reducer-like pattern (for example, using a single method to handle all color updates) instead of writing separate handlers for each color. Choose favorite color: 	K3	CO5	2.1.2 3.4.2 13.3.1